

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2457071.63114 +/- 0.00014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.133 +/- 0.0078
Transit depth (Rp/Rs)^2	1.77 +/- 0.21 [%]
Orbital Inclination (inc)	76.8 +/- 2.05
Airmass coefficient 1 (a1)	0.9515 +/- 0.0058
Airmass coefficient 2 (a2)	0.052 +/- 0.045
Scatter in the residuals of the lightcurve fit is	0.61 %
Best Comparison Star	#2 - [537, 252]
Optimal Aperture	4.34
Optimal Annulus	9.8
Transit Duration (day)	0.0727 +/- 0.0052

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.133 $\pm$ 0.0078 vs 0.1178 (deviation 1.88 σ)
Duration fit vs published	0.0727 d vs 0.125 d (deviation 9.72 σ)
Mid-time O-C	-1572.0 min
Depth SNR	16.5
Residual scatter	0.61 %

Light curve

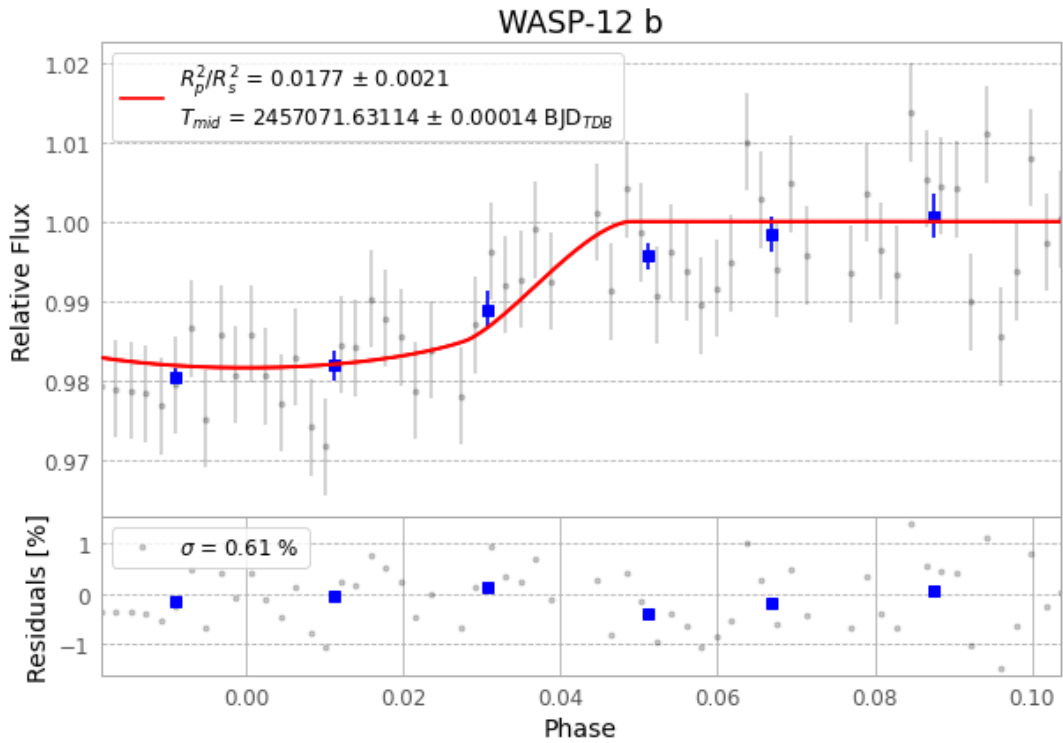


Figure 1 — FinalLightCurve_WASP-12 b_2015-02-17.png (SHA-256 ce4aa67fb5ac8fca...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [237, 269], 2 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 752059728214b6d8...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

63 FITS frames (41 MB), WASP-12150218023312.FITS → WASP-12150218054512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-150219.log (d3a3d0a4d582...), FinalLightCurve_WASP-12 b_2015-02-17.png (ce4aa67fb5ac...), FinalLightCurve_WASP-12 b_2015-02-17.csv (c630414ce44c...)

Compute resources

Wall time 90.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-07 15:24Z.